

Phase 1: Project Proposal

Group Number: 04

Group Name: TrackServ

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1. Purpose of the Project

- TrackServ is designed as an open, transparent, web-based platform to bridge the communication gap between local municipalities and the communities they serve.
- It operates as a two-sided civic engagement system that addresses both community and municipal needs.
- For citizens, it provides a public-facing portal to report civic issues—such as infrastructure damage or utility outages—and track their progress, addressing previous transparency gaps.
- For municipal staff, it provides a purpose-built administrative dashboard to triage, assign, and analyze the reports to improve operational efficiency.
- The platform also aims to improve data accuracy by utilizing integrated geolocation mapping to pinpoint the exact locations of reported issues.
- It features community voting and recognition (badges) to encourage sustained citizen engagement.

2. Implementation Language

- The frontend for both the citizen portal and the separate administrative dashboard was developed using JavaScript and React.
- The backend database relies on PostgreSQL, which is provided through the managed backend-as-a-service platform, Supabase.
- Originally, the proposal specified a custom stack using Node.js, PHP (for API development), and MySQL, but this was replaced during implementation in favor of Supabase's managed PostgreSQL infrastructure.

3. The SDLC Model

- The project followed a standard Software Development Life Cycle (SDLC) utilizing an iterative design approach known as the Spiral Model.
- This iterative approach enabled the development team to validate the core citizen reporting loop first.
- Once the core loop was validated, the team was able to add the administrative dashboard, staff assignment system, and analytics module as separate, and other functional add-ons later iterations.